THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A plant cell is placed in a solution and the cell contents shrink away from the wall. The solution is: 1

(a) hypotonic **(b)** isotonic **(c)** hypertonic **(d)** pure water

2. A cell has 8 chromosomes. After mitosis each daughter cell has: 1

(a) 4 **(b)** 2 **(c)** 16 **(d)** 8

3. The organelle which is found in all eukaryotic cells but whose number is highest in actively respiring cells: 1

(a) cell wall **(b)** nucleus **(c)** vacuole **(d)** mitochondria

4. Dividing cells of a root tip are best seen under the microscope in the: 1

(a) growing region **(b)** root hair zone **(c)** root cap **(d)** mature root

5. An animal cell placed in pure water will: 1

(a) shrink **(b)** swell and may burst
(c) divide **(d)** remain same

6. A cell with 8 chromosomes divides by meiosis. Each daughter cell has: 1

(a) 2 **(b)** 8 **(c)** 4 **(d)** 16

7. Which organelle is absent in animal cells? 1

(a) ribosome **(b)** chloroplast **(c)** Golgi **(d)** mitochondrion

8. The cell division that increases the number of cells for growth is: 1

(a) mitosis **(b)** fertilisation **(c)** osmosis **(d)** meiosis

9. Which cell structure would be most affected if a cell has no ribosomes? 1

(a) respiration only **(b)** photosynthesis
(c) protein synthesis **(d)** cell division

10. A cell in a solution neither gains nor loses weight. The solution is: 1

(a) isotonic **(b)** hypotonic **(c)** distilled water **(d)** hypertonic

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do red blood cells burst when placed in distilled water but not plant cells? 2

12. A cell has 12 chromosomes. How many are present in the cells formed by mitosis and meiosis? 2

13. State two differences between rough and smooth ER. 2

14. Why is a mature mammalian red blood cell without a nucleus advantageous? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A potato cube weighing 10 g is placed in distilled water and another in strong salt solution. Predict the masses after some time and explain. 3

16. Explain two major differences between a bacterial cell and an animal cell with the genetic material in each. 3

17. Explain why nerve cells are long and branched, using structure-function relation. 3

OR

Explain how the surface area to volume ratio limits the size of cells and why a big organism has many small cells. 3

18. The cell of a cow and that of a mouse are of similar sizes. How does the organism size differ? 3

OR

How does a unicellular organism like Amoeba perform the life functions without organs? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the structure and functions of the nucleus, including the nuclear membrane, nucleolus, chromatin and nuclear pores. 5

OR

Explain how the cell is organised as a miniature factory, giving the roles of the nucleus, ribosomes, ER, Golgi, mitochondria, lysosomes and the membrane. 5

20. Explain how the movement of substances across the cell membrane occurs: diffusion, osmosis and active transport with examples. 5

OR

Discuss the evidence that mitochondria and chloroplasts were once free-living bacteria and engulfed by early cells. 5

END OF QUESTION PAPER — Total: 40 Marks

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